

Michele Fumagalli

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Scientific Interests

Gas flows around galaxies, galaxy formation and evolution, the role of environment, absorption line systems, physics of the interstellar medium, star formation, stellar initial mass function.

Academic History

- 2020- **Professor**, *University of Milano Bicocca*.
- 2020- **Associate**, *INAF - Osservatorio Astronomico di Trieste*.
- 2020-2023 **Visiting Professor**, *Durham University*.
- 2018-2020 **Professor**, *Durham University*.
- 2017-2018 **Associate Professor (Reader)**, *Durham University*.
- 2014-2017 **Assistant Professor (Lecturer)**, *Durham University*.
- 2013-2014 **Postdoctoral Fellow**, *Carnegie Observatories, Princeton University*.

Education

- 2016 **Postgraduate Certificate in Academic Practice**, *Durham University, UK*.
- 2012 **Ph.D. in astrophysics**, *University of California, Santa Cruz, USA*.
- 2010 **Master in astrophysics**, *University of California, Santa Cruz, USA*.
- 2008 **Laurea specialistica (MSc)**, *University of Milano Bicocca, Italy*.
- 2006 **Laurea triennale (BSc)**, *University of Milano Bicocca, Italy*.

Selected Awards and Fellowships

- 2017 **Abilitazione Nazionale Italiana**, *Professore Associato e Ordinario*.
- 2015 **Fellow of the Higher Education Academy**.
- 2014-2015 **Carnegie Visiting Associate**, *Carnegie Observatories*.
Visiting fellowship at Carnegie Observatories.
- 2012 **Lyman Spitzer Fellowship**, *Princeton University*.
Postdoctoral fellowship in theoretical astrophysics.
- 2012 **Carnegie-Princeton Fellowship**, *Carnegie Observatories, Princeton University*.
Postdoctoral fellowship in observational astrophysics.

- 2012 **Hubble Fellowship**, Carnegie Observatories.
Awarded to highly qualified recent postdoctoral scientists to conduct independent research.
- 2012 **CfA Fellowship**, (declined), The Harvard-Smithsonian Center for Astrophysics.
Awarded to an outstanding researcher displaying significant promise in theory or observation.
- 2012 **Miller Research Fellowship**, (declined), University of California, Berkeley.
Awarded to exceptional young scientists of great promise.
- 2011 **Price Prize in Cosmology and AstroParticle Physics**, CCAPP, Ohio State University.
Awarded in recognition of research excellence in cosmology and astro-particle physics.
- 2011 **Chancellor's Dissertation Year Fellowship**, UCSC.
Awarded based on the academic achievement of the nominee.
- 2010 **Whitford Prize**, Department of Astronomy, UCSC.
Awarded for outstanding performance during the first and second years.
- 2008 **Regents' fellowship**, UCSC.
Awarded to promising first-year graduate students.

Grant History (principal investigator or primary co-investigator)

- 2023 **Bando Prin 2022 PNRR**, MUR, (Node PI).
- 2022 **Dipartimenti di Eccellenza 2023-2027**, MUR, (Deputy PI).
- 2020 **Durham Astronomy Consolidated Grant**, STFC, (Project co-PI).
- 2019 **NASA grant**, HST-GO-15637, (Science Co-PI).
- 2018 **ERC Attrattività**, Fondazione Cariplo, (PI).
- 2017 **ERC Starting Grant**, ERC, (PI).
- 2017 **Durham Astronomy Consolidated Grant**, STFC, (Project PI).
- 2015 **NASA grant**, HST-GO-14127, (Science PI).
- 2012 **NASA Hubble Fellowship**, grant HF-51305.01-A, (PI).
- 2010 **HIPACC grant**, University California, (PI).

Talks, Seminars, Conferences

- Oct., 2025 **Multi-phase, Multi-temperature and Complex**, *Olbia, Italy*, Invited review.
The multiphase nature of the CGM: warm ionized gas in absorption and emission.
- Aug., 2025 **Ascona meeting on cosmic ISM and Stellar Populations**, *Ascona, Switzerland*, Invited.
Mapping the cosmic baryon cycle over 10 billion years in the MUSE era
- Nov., 2024 **A decade of discoveries with MUSE and beyond**, *ESO*, Invited Review.
The MUSE contribution to mapping the cosmic baryon cycle over 10 billion years
- Sep., 2024 **ILR workshop at Osaka**, *Osaka, Japan*, Invited.
The gas environment of galaxies across 10 billion years
- Jun., 2024 **IFPU Focus Week: Where the CircumGalactic medium meets the galaxy environment**, *Italy*, SOC member.
- May, 2024 **OPINAS Colloquium**, *Max-Planck for Extraterrestrial Physics*, invited.
The gas environment of galaxies across 10 billion years
- Apr., 2024 **Extreme Galaxies in their Extreme Environments at Extremely Early Epochs**, *Iceland*, SOC member.
- Mar., 2024 **Bologna Joint Astrophysical Colloquium**, *INAF/University of Bologna, IT*, invited.
The gas environment of galaxies across 10 billion years

- Jan., 2024 **Astronomy colloquium, Lancaster, UK**, invited.
The gas environment of galaxies across 10 billion years
- Sep., 2023 **IMAGING 2023, Italy**, Organizing committee.
Connecting the dots between the CGM and the larger-scale environment
- May., 2023 **IASF Colloquium Series, IASF-INAF, Milan**, Invited.
The gas environment of galaxies across 10 billion years
- Apr., 2023 **The Department of Astronomy Colloquium Series, Tsinghua University**, Invited.
The gas environment of galaxies across 10 billion years
- Mar., 2023 **The Circum-Galactic Medium across cosmic time: an observational and modeling challenge, The 52nd "Saas-Fee Advanced Course"**, Invited lecturer.
The multiphase CGM in absorption and emission
- Feb., 2023 **The Multiphase Circumgalactic Medium, Ringberg Castle, Germany**, Invited.
The gas environment of galaxies across 10 billion years
- Nov., 2022 **IoA colloquium, Cambridge, UK**, invited.
The gas environment of galaxies across 10 billion years
- Sep., 2022 **What Matter(s) Around Galaxies 2022, Italy**, SOC chair and lead organizer.
Connecting the dots between the CGM and the larger-scale environment
- Jun., 2021 **KIAA Forum on Gas in Galaxies for Early Career Scientists, Kavli Institute for Astronomy and Astrophysics**, invited.
Flows around galaxies: advancements, challenges and opportunities
- May., 2021 **Multi-object Spectroscopy for Statistical Measures of Galaxy Evolution Workshop, Space Telescope Science Institute**, invited review.
Studying gas flows around galaxies with multi object spectroscopy
- May., 2021 **Extragalactic Seminar Series, University of Victoria**, invited.
Shedding light on gas around galaxies across cosmic times
- Apr., 2021 **Astronomy Colloquium, University of California, Santa Cruz**, invited.
Shedding light on gas around galaxies across cosmic times
- Mar., 2021 **Physics Colloquium, North Carolina State University**, invited.
Shedding light on gas around galaxies across cosmic times
- Nov., 2020 **Kapteyn Institute Colloquium, University of Groningen**, invited.
Shedding light on gas around galaxies across cosmic times
- Jun., 2020 **EAS 2020, Leiden**, invited review.
Flows around galaxies in 2020: advancements, challenges and opportunities
- Jun., 2020 **Insights into the CGM and ICM, IAP, France**, invited.
MUSE observations of the CGM of distant galaxies
- Apr., 2020 **Astronomy Colloquium, Royal Observatory/Edinburgh**, invited.
Shedding light on gas around galaxies across cosmic times
- Mar., 2020 **Astronomy Colloquium, INAF/Arcetri**, invited.
Shedding light on gas around galaxies across cosmic times
- Dec., 2019 **Joint Astronomy Colloquium, MPA-MPE-ESO**, invited.
Shedding light on gas around galaxies across cosmic times
- Oct., 2019 **CGM in Berlin 2019, Max Planck Society**, invited.
Gas around galaxies at $z \sim 2 - 3$: linking emission and absorption with large surveys
- Jun., 2019 **What Matter(s) Between Galaxies, Abbazia di Spineto, SOC**.
Gas around galaxies: connecting emission and absorption with large surveys
- Mar., 2019 **Astronomy Seminar, Nottingham University**, invited.
Shedding light on gas around galaxies across cosmic time

- Dec., 2018 **Twenty years of science at Bicocca**, *Milano-Bicocca University*, invited review.
Astrophysics ± 20 : Deeper, Sharper, and Bigger
- Nov., 2018 **CASTOR UV space observatory**, *The Royal Observatory Edinburgh*, invited review.
The galaxy-IGM connection
- Jun., 2017 **What Matter(s) Around Galaxies**, *Durham University*, SOC/LOC co-chair.
Probing the gaseous environment of star-forming galaxies in absorption and emission
- Apr., 2017 **Seminar, Department of Physics, University of Milano-Bicocca**, invited.
MUS(E)ing over gas flows as drivers of galaxy evolution
- May., 2016 **Cavendish Astrophysics Seminar, University of Cambridge**, invited.
Gas flows as fuel for star formation: a spotlight on strong absorption line systems
- Apr., 2016 **Astronomy Seminar, ETH Zurich**, invited.
Gas flows as fuel for star formation: a spotlight on strong absorption line systems
- Mar., 2016 **Astronomy Seminar, Stockholm University**, invited.
Gas flows as fuel for star formation: a spotlight on strong absorption line systems
- Sep., 2015 **Astronomy Seminar, INAF/Trieste**, invited.
Gas flows as fuel for star formation: a spotlight on strong absorption line systems
- Jun., 2015 **IGM@50, INAF/Firenze**, invited.
Probing gas flows near galaxies: a spotlight on Lyman Limit Systems
- Jun., 2014 **Intergalactic Matters, MPIA, Heidelberg**, invited.
A shot in the dark: the star formation rates of DLAs at $z \sim 2 - 3$
- Apr., 2014 **Colorful galaxies: a conference for Peppo Gavazzi's birthday, Como, Italy**, invited.
Can we use H α to trace star formation rates?
- Apr., 2014 **Exploiting VST ATLAS... and its sister surveys, Durham University**, invited.
ATLAS search for Lyman Limit Systems in quasar pairs.
- Mar., 2014 **Astronomy Friday Lunch Talks, Durham University**.
The importance of stochastic effects in stellar population synthesis.
- Jan., 2014 **DEX meeting, Durham University**.
Investigations on the gaseous environment of distant galaxies.
- Dec., 2013 **TAPIR seminar, Caltech**, invited.
Investigations on the gaseous environment of distant galaxies.
- Oct., 2013 **Metal Production and Distribution in a Hierarchical Universe, Rencontres de l'Observatoire de Paris 2013 - ESO Workshop**, invited review.
IGM abundances in the high-redshift universe.
- Aug., 2013 **Santa Cruz Galaxy Workshop, UCSC**.
Lyman limit systems and the circumgalactic medium at $z \sim 2 - 3$.
- Jun., 2013 **Intergalactic Interactions, Higgs Centre, Edinburgh**, invited.
Lyman limit systems and the circumgalactic medium at $z \sim 2 - 3$.
- Jun., 2013 **ENIGMA workshop, MPIA**, invited.
Lyman limit systems and the circumgalactic medium at $z \sim 2 - 3$.
- Apr., 2013 **Lunch Talk, Carnegie Observatories**.
Beyond the disk: The role of halo gas in galaxy formation.
- Mar., 2013 **Hubble Fellows Symposium, STScI, Baltimore**.
Optically-thick hydrogen in the $z=3$ universe
- Dec., 2012 **University of Milano-Bicocca, Milan**, invited.
The gaseous environment of distant galaxies
- Nov., 2012 **UT Astronomy Colloquium, Austin**, invited.
The gaseous environment of distant galaxies

- Sep., 2012 **Keck Science Meeting**, *San Diego*.
Pristine gas two billion years after the Big Bang
- Jun., 2012 **Metals in Tuscany**, *INAF/Firenze*, invited.
Pristine gas two billion years after the Big Bang
- May., 2012 **Price Prize lecture**, *CCAPP Ohio State University*, invited.
Cosmology with absorption line systems
- Apr., 2012 **Astronomy Colloquium**, *Osservatorio Astronomico di Brera*, invited.
Cosmology with absorption line systems
- Mar., 2012 **Turbulence in Cosmic Structure Formation**, *Arizona State University*.
Detection of pristine gas two billion years after the Big Bang
- Jan., 2012 **DARK Cake Meeting**, *DARK Cosmology Centre*.
Detecting cold accretion and metal poor gas around galaxies
- Jan., 2012 **219th AAS Meeting**, *Austin, TX*.
Exploring the gas cycle in high-redshift galaxies: a joint effort of theory and observations
- Dec., 2011 **Theory meeting of the Galaxy and Cosmology group**, *MPIA Heidelberg*.
Probing inflow in high-redshift galaxies
- Oct., 2011 **Theoretical Astrophysics Center seminar**, *UC Berkeley*, invited.
Exploring the gas cycle in high-redshift galaxies: a joint effort of theory and observations
- Oct., 2011 **Lunch Talk**, *Carnegie Observatories*.
Exploring the gas cycle in high-redshift galaxies: a joint effort of theory and observations
- Oct., 2011 **Astronomy Tea Talk**, *Caltech*.
Exploring the gas cycle in high-redshift galaxies: a joint effort of theory and observations
- Aug., 2011 **Santa Cruz galaxy workshop**, *Santa Cruz*.
Cold streams and primordial gas at high redshift
- Jul., 2011 **Celebrating the career of A. Wolfe**, *Schloss Ringberg*, invited.
Detecting cold streams with absorption line systems
- Jul., 2011 **MPIA**, *Heidelberg*.
Stochastic star formation and IMF (non) variation
- Jun., 2011 **Odyssey of cosmic baryons**, *Marseille*.
Detecting cold streams with absorption line systems
- Jun., 2011 **Gas in galaxies**, *Kloster Seeon, Germany*.
Detecting cold streams with absorption line systems
- Dec., 2010 **CASS, UCSD**, *San Diego*.
Gas in and around galaxies
- Aug., 2010 **Santa Cruz galaxy workshop**, *Santa Cruz*.
Gas in simulations of $z > 2$ galaxies
- May, 2010 **Como+Milano+Heidelberg+Marseille**.
Images and simulations to connect gas and stars in $z > 2$ galaxies
- Apr., 2010 **UCSC**, *Santa Cruz*.
Hunting gas and stars in galaxies across the Universe
- Aug., 2009 **Santa Cruz Galaxy Workshop**, *Santa Cruz*.
A shot in the dark: probing galaxies giving rise to DLAs at $z > 2$
- Aug., 2009 **UCSC Friday Lunch Talk**, *Santa Cruz*.
Molecular gas deficiency in HI poor galaxies
- Jun., 2009 **University of Chicago**, *Chicago*.
A shot in the dark: imaging of DLAs
- Mar., 2009 **Università dell'Insubria**, *Como, Italy*.
Star formation $z = 0 - 3$

- Dec., 2008 **CASS, UCSD, San Diego.**
The star formation rate and gas content in local spiral galaxies
- Jul., 2008 **Università di Milano-Bicocca, Milano, Italy.**
The relationship between gas content and star formation rate in spiral galaxies

Proposal History (principal investigator or primary co-investigator)

- 2026 JWST; 54 hours, cycle 5.
- 2024 ALMA; 32 hours, cycle 11.
- 2023 HST; 100 SNAP targets, cycle 31.
- 2022 ALMA; 9 hours, cycle 9.
- 2022 ESO/VLT; 16 hours, P109.
- 2021 ALMA; 9 hours, cycle 8.
- 2019 ESO/VLT; 25 hours, P105.
- 2019 Hubble Space Telescope; 8 orbits, cycle 27.
- 2019 JCMT/SCUBA-2; 30 hours, 2019B.
- 2018 Hubble Space Telescope; 90 orbits, cycle 26 (LP).
- 2017 ESO/VLT; 250 hours, P101 (LP).
- 2018 JCMT/SCUBA-2; 16 hours, 2018A.
- 2017 ESO/VLT; 36 hours, P100.
- 2017 JCMT/SCUBA-2; 9 hours, 2017B.
- 2016 ESO/VLT; 18 hours, P99.
- 2016 Keck Telescope; 2 nights, 2016B.
- 2016 Hubble Space Telescope; 96 orbits, cycle 24 (LP).
- 2016 JCMT/SCUBA-2; 9 hours, 2016B.
- 2016 Keck Telescope; 1 night, 2016A.
- 2016 WHT; 12 nights, 2016A.
- 2016 ESO/VLT; 106 hours, P97-100 (LP).
- 2015 WHT; 9 nights, 2015B.
- 2015 ESO/VLT; 9 hours, P96.
- 2015 Hubble Space Telescope; 55 orbits, cycle 23.
- 2014 ESO/VLT; 28 hours, P95.
- 2014 ESO/VLT; 5 hours, P94.
- 2014 Gemini-S Telescope; 30 hours, 2014A.
- 2014 Magellan Telescope; 4 nights, 2014A.
- 2013 Magellan Telescope; 5 nights, 2013B.
- 2013 Keck Telescope; 1 night, 2013B.
- 2012 Keck Telescope; 1 night, 2013A.
- 2012 Magellan Telescope; 4 nights, 2013A.
- 2012 Magellan Telescope; 4 nights, 2012B.
- 2011 IRAM 30m Telescope; 64 hours, 2011B.

Teaching and Advising

- 2024- Introduction to Galaxies; MSc at University of Milano-Bicocca.
 - 2024- Didactics of Mathematics; School of Education, University of Milano-Bicocca.
 - 2021-2026 Medical Physics; School of Medicine, University of Milano-Bicocca.
 - 2019- Astrophysics Laboratory; MSc at University of Milano-Bicocca.
 - 2018-2019 Radiative processes in astrophysics; PhD lecture series at Durham University.
 - 2018 The role of baryonic process in galaxy formation and evolution; PhD lecture series at University of Milano-Bicocca.
 - 2016-2017 PHYS2651: Physics in Society, BSc at Durham University.
 - 2014-2019 PHYS1081: Introduction to Astronomy, BSc at Durham University.
 - 2014-2018 PHYS1101: Discovery Skills in Physics, BSc at Durham University.
 - 2009 Ay2: Overview of the Universe, BSc at UCSC.
- PhD Students
- Mr. Michele Fulghieri (2025-), University of Milano-Bicocca.
 - Mr. Davide Tornotti (2024-), University of Milano-Bicocca.
 - Mr. Georg Herzog (2020-2024), University of Milano-Bicocca (PhD, 2024).
 - Mr. Calvin Sykes (2017-2021), Durham University (PhD, 2021).
 - Ms. Louise Welsh (2017-2021), Durham University (PhD, 2021).
 - Mr. Ruari Mackenzie (2014-2018), Durham University (PhD, 2018).
 - Mr. Greg Ashworth (2014-2018), Durham University (PhD, 2018).
- PDRAs
- Dr. Francesco Pistis (2023-2026), University of Milano-Bicocca.
 - Dr. Trystyn Berg (2021-2023), University of Milano-Bicocca.
 - Dr. Louise Welsh (2021-2023), University of Milano-Bicocca.
 - Dr. Alessia Longobardi (2021-2023), University of Milano-Bicocca.
 - Dr. Emma Lofthouse (2018-2023), Durham University, University of Milano-Bicocca.
 - Dr. Rajeshwari Dutta (2019-2022), Durham University, University of Milano-Bicocca.
 - Dr. Alejandro Benitez Llambay (2021-2022), University of Milano-Bicocca.
 - Dr. Matteo Fossati (2018-2021), Durham University, University of Milano-Bicocca.
 - Dr. Elisabeta Lusso (2017-2019), Junior Research Fellow, Durham University.
 - Dr. Richard Bielby (2017-2019), Durham University.

Membership and Activities

- 2026- Spokesperson for Public Engagement, Physics Department, University of Milano-Bicocca
- 2026- Associate Member (elected), Sigma Xi, The Scientific Research Honour Society
- 2025- Tiger-team co-lead and member, wide-field spectroscopic telescope (WST) consortium
- 2023-2026 Co-lead of the outreach project “Un nuovo sguardo sul cielo di Milano” funded by NextGenerationEU
- 2023- Steering Committee, Bicocca Centre for Quantitative Cosmology, Dipartimenti di Eccellenza 2023

- 2022- Member of the International Astronomical Union
- 2021- Member of the Euclid Consortium
- 2021- Member of the MOSAIC/ELT Science working groups “First Light” and “Inventory of matter”
- 2021- Member of the Science Working Group, WEAVE survey
- 2020- Coordinator of Absorption Line Studies in the Quasar Working Group, WEAVE survey
- 2020,2025 Chair of PhD Admission Committee, Physics Department, University of Milano-Bicocca
- 2020 Panel Member, USA National Science Foundation
- 2018- Peer reviewer, Nature
- 2018- Peer reviewer, European Research Council
- 2017- Peer reviewer, Nature Astronomy
- 2016-2018 Member of Van Mildert College Council, Durham University
- 2016- HIRES/ELT Galaxy and IGM Working Group
- 2012- Peer reviewer, Astrophysical Journal
- 2012- Peer reviewer, Monthly Notices of the Royal Astronomical Society
- 2012- Peer reviewer, Astronomy and Astrophysics
- 2011-2012 Graduate Student Mentor, UCSC Astronomy & Astrophysics Department
- 2011-2015 Member, European Physical Society
- 2011-2012 Member, American Astronomical Society
- 2008-2015 Member, Società Italiana di Fisica

Refereed publications

1. Wisz, M., et al. 2026, ApJ in press, arXiv:2604.16751. *The Qz5 Survey (II): Metallicity Evolution of Damped Ly α Systems Out to $z \sim 5$.*
2. Euclid Collaboration. 2026, arXiv:2601.10709. *Euclid preparation. 3D reconstruction of the cosmic web with simulated Euclid Deep spectroscopic samples.*
3. Galbiati, M., Tornotti, D., **Fumagalli, M.**, et al. 2025, arXiv:2512.16422. *MUSE Analysis of Gas around Galaxies (MAGG) - VII. Emission line galaxies near strong blended Ly α absorption systems at $z \gtrsim 3$.*
4. Chaudhary, R., Joshi, R., Das, S., **Fumagalli, M.** et al. 2025, A&A submitted (arXiv:2509.02672). *Baryonic Ecosystem IN Galaxies (BEINGMgII) – III. Cool gas reservoirs at $0.3 \leq z \leq 1.6$ in the Dark Energy Survey.*
5. Santo, E., **Fumagalli, M.**, Chang, S-K, et al. 2026, A&A, 707, A359. *The MUSE Ultra Deep Field (MUDF) VIII. The cool gas distribution surrounding galaxies at redshifts $z \sim 0.5 - 2$.*
6. Scialpi, M. et al. 2026, A&A 709, A236. *Cosmic Duets I. High-spatial resolution spectroscopy of dual and lensed AGN with MUSE.*
7. Tornotti, D., Fossati, M., **Fumagalli, M.**, et al. 2025, A&A, 704, 201. *Bayesian luminosity function estimation in multidepth datasets with selection effects: a case study for $3 < z < 5$ Ly α emitters.*
8. **Fumagalli, M.** 2024, arXiv:2409.00174. *The multiphase circumgalactic medium and its relation to galaxies: an observational perspective.*
9. Euclid Collaboration. 2025, A&A, 697, 1. *Euclid. I. Overview of the Euclid mission.*

10. Das, S., et al. 2025, A&A, 695, 207. *Baryonic Ecosystem in Galaxies (BEINGMgII) – II. Unveiling the Nature of Galaxies Harboring Cool Gas Reservoirs.*
11. Oyarzun, G.-A., et al. 2025, ApJ, 983, 10. *The Qz5 Survey (I): How the HI Mass Density of the Universe Evolves With Cosmic Time.*
12. Joshi, R., Das, S., **Fumagalli, M.** et al. 2025, A&A, 695, 206. *Baryonic Ecosystem in Galaxies (BEINGMgII). Host Galaxies of Ultra-strong MgII Absorbers in Subaru Hyper Suprime-Cam Survey.*
13. Travascio, A., et al. 2025, A&A, 694, 165. *X-ray view of a massive node of the Cosmic Web at $z \sim 3$ I. An exceptional overdensity of rapidly accreting SMBHs.*
14. Pensabene, A., Galbiati, M., **Fumagalli, M.** et al. 2025, A&A, 696, 33. *The MUSE Ultra Deep Field (MUDF) VII. Probing high-redshift gas structures in the surroundings of ALMA-identified massive dusty galaxies.*
15. Galbiati, M., et al. 2025, A&A, 696, 95. *Connecting the growth of galaxies to the large-scale environment in a massive node of the Cosmic Web at $z \sim 3$.*
16. Pistis, F., **Fumagalli, M.**, Fossati, M., et al. 2025, A&A, 698, 292. *Automated quasar continuum estimation using neural networks: a comparative study of deep-learning architectures.*
17. Wang, W., et al. 2025, Nature Astronomy, 9, 710. *A Giant Disk Galaxy Two Billion Years After The Big Bang.*
18. Mizener, A., et al. 2025, ApJ, 989, 114. *CLUES III: Do User Choices Impact The Results of SED Fitting? Tests of 'Off-The-Shelf' Stellar Population and Dust Extinction Models in the CLUES Sample.*
19. Chang, S.-J., et al. 2025, MNRAS, 542, 525. *Modeling Mg II resonance doublet spectra from galaxy haloes at $z \sim 1$.*
20. Tornotti, D., **Fumagalli, M.**, Fossati, M., et al. 2025, ApJL, 980, 43. *The MUSE Ultra Deep Field: A 5 Mpc stretch of the $z \approx 4$ cosmic web revealed in emission.*
21. Tornotti, D., **Fumagalli, M.**, Fossati, M., et al. 2025, Nat. Ast., 9, 577. *High-definition imaging of an extended filament connecting active quasars at cosmic noon.*
22. Galbiati, M., Dutta, R., **Fumagalli, M.** et al. 2024, A&A, 690, 7. *MUSE Analysis of Gas around Galaxies (MAGG) – VI. The cool and enriched gas environment of $z \geq 3$ Ly α emitters.*
23. Benitez-Llambay, A., Dutta, R., **Fumagalli, M.**, Navarro, J.F.. 2024, ApJ, 973, 61. *Examining the Nature of the Starless Dark Matter Halo Candidate Cloud-9 with Very Large Array Observations.*
24. D'Odorico, V., et al. 2024, Experimental Astronomy, 58, 21. *Galaxy Formation and Symbiotic Evolution with the Inter-Galactic Medium in the Age of ELT-ANDES.*
25. Dharmender, Joshi, R., **Fumagalli, M.** et al. 2024, A&A Letters, 692, 7. *Star-Formation in Neutral Hydrogen Gas Reservoirs at Cosmic Noon.*
26. Welsh, L., Cooke, R., **Fumagalli, M.**, et al. 2024, 691, 285. *A survey of extremely metal-poor gas at cosmic noon: evidence of elevated [O/Fe].*
27. Arnaudova, M.I. et al. 2024, A&A, 535, 2269. *WEAVE First Light Observations: Origin and Dynamics of the Shock Front in Stephan's Quintet.*
28. Dutta, R., **Fumagalli, M.**, Fossati, M., et al. 2024, 691, 236. *Metal line emission around $z < 1$ galaxies.*

29. Pruto, G., **Fumagalli, M.**, Rafelski, M., et al. 2024, 691, 222. *The stellar population of a $z \approx 3.25$ Ly α emitting group associated with a damped Ly α absorber.*
30. Revalski, M., et al. 2024, ApJ, 966, 228. *The MUSE Ultra Deep Field (MUDF). V. Characterizing the Mass-Metallicity Relation for Low Mass Galaxies at $z \approx 1 - 2$.*
31. Beckett, A. et al. 2024, ApJ, 974, 256. *The MUSE Ultra Deep Field (MUDF). VI. The relationship between galaxy properties and metals in the circumgalactic medium.*
32. Pensabene, A, et al. 2024, A&A, 684, 119. *ALMA survey of a massive node of the Cosmic Web at $z \sim 3$. I. Discovery of a large overdensity of CO emitters.*
33. Siressi, M., et al. 2024, AJ, 167, 166. *CLusters in the Uv as EngineS (CLUES). II. Sub-kpc scale outflows driven by stellar feedback.*
34. Dutta, R., Acebron, A., **Fumagalli, M.** et al. 2024, MNRAS, 528, 1895. *Probing coherence in metal absorption towards multiple images of strong gravitationally lensed quasars.*
35. Finn, M.K., et al. 2024, ApJ, 964, 13. *ALMA-LEGUS II: The Influence of Sub-Galactic Environment on Molecular Cloud Properties.*
36. Finn, M.K., et al. 2024, ApJ, 964, 12. *ALMA-LEGUS I: The Influence of Galaxy Morphology on Molecular Cloud Properties.*
37. Bortolini, G., et al. 2024, MNRAS, 527, 5339. *The spatially resolved star formation history of the dwarf spiral galaxy NGC 5474.*
38. Jin, S., et al. 2024, MNRAS, 530, 2688 *The wide-field, multiplexed, spectroscopic facility WEAVE: Survey design, overview, and simulated implementation.*
39. Stephenson, H.M.O, et al. 2023, MNRAS, 527, 7891. *Quasar Sightline and Galaxy Evolution (QSAGE) – III. The mass-metallicity and fundamental metallicity relation in $z \sim 2.2$ galaxies.*
40. de Beer, S. et al. 2023, MNRAS, 526, 1850 *Resolving the physics of Quasar Ly α Nebulae (RePhyNe): I. Constraining Quasar host halo masses through Circumgalactic Medium kinematics.*
41. Lusso, E., Nardini, E., **Fumagalli, M.** et al. 2023, MNRAS, 525, 4388. *The MUSE Ultra Deep Field (MUDF). IV. A pair of X-ray weak quasars at the heart of two extended Ly α nebulae.*
42. Jung, D.E. et al. 2023, ApJ, 954, 136. *Universal Upper End of the Stellar Initial Mass Function in the Young and Compact LEGUS clusters.*
43. Longobardi, A., Fossati, M., **Fumagalli, M.** et al. 2023, RASTI, 2, 470. *Towards an automatic approach to modelling the circumgalactic medium: new tools for mock making and fitting of metal profiles in large surveys.*
44. Welsh, L., Cooke, R., **Fumagalli, M.**, Pettini, M. 2023, MNRAS, 525, 527. *Towards ultra metal-poor DLAs: linking the chemistry of the most metal-poor DLA to the first stars.*
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5. **Fumagalli, M.** 2014, MmSAI, 85, 355. *Metal abundances in the high-redshift intergalactic medium.*
6. **Fumagalli, M.** 2012, Ph.D. dissertation, University of California, Santa Cruz. *Food for stars: the role of hydrogen in the formation and evolution of galaxies.*
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